SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Commercial Livestock Trading Platform

Revised Phase 1 Requirements

WIL Project Phase 2

Flutter • ASP.NET • Supabase

1. Project Introduction

The proposed project is a Commercial Livestock Trading Platform. Phase 2 continues the revised Phase 1 requirements and expands the platform from livestock discovery and filtering into a complete buyer-to-seller trading workflow.

The platform will continue to use one stakeholder as the only seller while supporting buyers and multiple livestock categories. Buyers can browse and filter listings, view detailed livestock information, make offers, place orders, provide delivery information, make payments, track orders, confirm receipt, and rate/review completed orders. The stakeholder can manage listings, respond to offers, manage delivery details, and view a financial summary.

The platform remains focused on the local and national market. International buyers, international payments, currency conversion and international delivery remain outside the project scope.

2. Stakeholder Profile and Current Process

2.1 Organisation and Stakeholder Context

The stakeholder operates in the commercial livestock sector and is the only seller on the proposed platform. The main business need remains improved access to buyers through a centralized digital platform, while Phase 2 adds transactional functionality to support the process from livestock discovery through completed purchase.

2.2 Stakeholder / Seller

- Create, update and remove livestock listings.
- Enter and manage livestock details such as type, price and weight.
- Provide brandmark information and upload ownership documentation.
- Provide or manage vaccination information where required.
- View buyer offers and accept or reject offers.
- Manage relevant delivery details for accepted/placed orders.
- View a financial summary of sales and related transactions.

2.3 Buyers

- Register and log in.
- Browse and filter livestock listings.
- View detailed livestock information, including ownership/brandmark and relevant vaccination information where provided.
- Make an offer on livestock.
- Place an order after an offer/purchase is accepted or otherwise permitted by the business rules.
- Provide a delivery address.
- Make a payment through the approved payment process.
- Track an order.
- Confirm receipt of livestock.
- Rate and review a completed order.

2.4 Existing Process

The platform addresses the stakeholder's need for improved market access by centralising livestock listings and extending the digital process to offers, orders, payment, delivery tracking and post-purchase feedback.

3. Problem Statement

The platform must address both the discovery problem identified in Phase 1 and the lack of an integrated digital process for progressing buyer interest into an offer, accepted transaction, order, payment, delivery and post-purchase review.

Identified Problems

1. The stakeholder is struggling to reach potential buyers.
2. Local buyers need an easier way to find the stakeholder's livestock.
3. Buyers need to compare livestock using important information such as price, type and weight.
4. Buyers need a filtering function so that they do not have to look through every listing.
5. Buyers need access to ownership and brandmark information where it is provided.
6. The system needs to support multiple livestock types while keeping the first implementation focused on the core commercial trading functions.

3.1 Current Situation

The stakeholder currently needs a better way to present livestock to potential buyers. Buyers may also find it difficult to identify livestock that meet their requirements when information is not presented in one central livestock trading platform.

3.2 Problem

The main problem is the lack of a commercial livestock trading platform that allows the stakeholder to advertise livestock and allows buyers to easily find and filter livestock according to their needs.

3.3 Affected Users

- The stakeholder as the only seller.
- Local and national buyers.

3.4 Consequences

Without a suitable livestock trading platform, the stakeholder may have fewer opportunities to reach potential buyers. Buyers may also spend more time searching for suitable livestock and comparing information such as price, type and weight.

3.5 Need for Improvement

A single platform is required to make livestock easier to discover while providing a structured and traceable digital trading workflow for offers, orders, payment, delivery and customer feedback.

4. Project Aim and Objectives

4.1 Project Aim

The aim is to develop a commercial livestock trading platform that allows the stakeholder to advertise and manage livestock and allows buyers to discover, evaluate and purchase livestock through an integrated digital trading workflow.

4.2 Project Objectives

- Support multiple livestock categories.
- Allow the stakeholder to create, update and remove livestock listings.
- Allow buyers to browse and filter livestock by relevant listing information.
- Display detailed livestock information, including price, type, weight and available ownership, brandmark and vaccination information.
- Allow buyers to make offers on livestock.
- Allow the stakeholder to accept or reject buyer offers.
- Support order placement and delivery-address capture.
- Support payment for approved orders.
- Allow buyers to track orders and confirm receipt.
- Allow buyers to rate and review completed orders.
- Allow the stakeholder to manage delivery details.
- Provide the stakeholder with a financial summary of sales/transactions.
- Maintain secure role-based access and protect buyer, seller and transaction information.

5. Project Scope

5.1 In-Scope

Area	Phase 2 Scope
User Management	Buyer registration/login; stakeholder/seller login; role-based access.
Livestock Trading platform	Multiple categories; listing creation, update and removal; browsing; detailed livestock information; filtering.
Livestock Information	Type, price, weight and relevant date-of-birth, parent and vaccination information where required.
Ownership & Identification	Ownership documentation upload; brandmark information; appropriate display to buyers.
Offers	Buyer makes an offer; stakeholder views and accepts/rejects offers.
Orders	Order creation/placement for an eligible transaction; order status management.
Delivery	Buyer delivery address; stakeholder delivery-detail updates; buyer order tracking.

Payments	Payment for approved orders through the selected/approved payment mechanism.
Completion	Buyer confirms receipt of livestock.
Ratings & Reviews	Buyer rates and reviews completed orders.
Seller Finance	Stakeholder can view a financial summary of sales/transactions.
Technology Integration	Flutter application, ASP.NET backend/API and Supabase services/database.

5.2 Deliverables

- Functional Commercial Livestock Trading Platform.
- Buyer functionality covering discovery through order completion and review.
- Stakeholder/seller functionality covering listing and transaction management.
- Database supporting users, livestock, offers, orders, delivery, payments and reviews.
- Integrated Flutter, ASP.NET and Supabase application.
- Updated system architecture, use cases, sequence diagrams and ERD aligned with these requirements.

5.3 Acceptance Criteria

- A buyer can register and log in.
- The stakeholder can log in as the only seller.
- The stakeholder can create, update and remove listings.
- Buyers can browse and filter listings by available criteria.
- Buyers can view relevant livestock details, including ownership/brandmark and vaccination information where provided.
- A buyer can submit an offer.
- The stakeholder can accept or reject an offer.
- An eligible accepted transaction can result in an order.
- The buyer can provide a delivery address.
- The buyer can complete an approved payment process.
- The buyer can track order status.
- The buyer can confirm receipt.
- The buyer can rate and review a completed order.
- The stakeholder can update delivery details.
- The stakeholder can view a financial summary.
- Information is stored and retrieved correctly and protected according to user role.

5.4 Exclusions

- Multiple sellers or farmer accounts beyond the single stakeholder.
- International buyers/markets, international payments, currency conversion and international delivery.
- Formal verification of ownership documents.
- Formal verification of brandmarks.
- Insurance services.
- Advanced auction functionality unless separately approved. The Phase 2 offer mechanism is a direct buyer offer that the stakeholder can accept or reject.

5.5 Constraints

- Development time and team resources.
- Technical skills and experience of the team.
- Dependence on selected backend, database, authentication, storage, payment and delivery services.
- Payment and delivery functionality must follow the capabilities of the selected services.
- Requirements may change following further stakeholder consultation.
- The project must protect buyer, livestock, ownership and transaction information.

5.6 Assumptions

- Buyers are the main trading platform users.
- Multiple livestock categories are supported.
- The stakeholder provides accurate livestock and ownership information.
- Buyers provide accurate delivery information.
- The selected payment mechanism is available and suitable for the project.
- Delivery tracking information can be supplied or updated through the agreed delivery process.

6. Stakeholder Needs

Stakeholder Need	Priority
Improve access to the livestock market.	High
Advertise and manage livestock commercially.	High
Allow buyers to browse and filter livestock.	High
Provide detailed livestock and ownership information.	High
Allow buyers to make offers.	High
Allow the stakeholder to accept/reject offers.	High
Support orders, delivery address and payment.	High
Allow order tracking and receipt confirmation.	High
Allow ratings and reviews.	Medium
Provide delivery-detail management.	High
Provide a financial summary to the stakeholder.	Medium
Support multiple livestock categories.	High

7. Functional Requirements

7.1 User Registration and Access

- FR-01: The app shall allow buyers to register for an account.
- FR-02: The app shall allow the stakeholder to access the system as the seller.
- FR-03: The app shall provide secure role-based access.

7.2 Stakeholder / Seller Listing Functionality

- FR-04: The app shall allow the stakeholder to create livestock listings.

- FR-05: The stakeholder shall enter livestock price.
- FR-06: The stakeholder shall enter livestock type/category.
- FR-07: The stakeholder shall enter livestock weight.
- FR-08: The stakeholder shall provide brandmark information.
- FR-09: The stakeholder shall upload ownership documentation.
- FR-10: The stakeholder shall update livestock listing information.
- FR-11: The stakeholder shall remove livestock listings.
- FR-12: The stakeholder shall manage relevant livestock information.

7.3 Livestock Information

- FR-13: The stakeholder shall record date of birth.
- FR-14: The stakeholder shall record parent information.
- FR-15: The stakeholder shall record vaccination information.
- FR-16: The system shall store and retrieve relevant livestock information.
- FR-17: Buyers shall be able to view relevant livestock details provided for a listing.

7.4 Buyer Functionality

- FR-18: Buyers shall be able to browse available livestock listings.
- FR-19: Buyers shall be able to view detailed livestock information.
- FR-20: Buyers shall be able to filter listings using available criteria.
- FR-21: Buyers shall be able to filter by price or price range.
- FR-22: Buyers shall be able to filter by livestock type/category.
- FR-23: Buyers shall be able to filter by weight or weight range.
- FR-24: The displayed listings shall update according to selected filters.

7.5 Offers

- FR-25: Buyers shall be able to make an offer on an eligible livestock listing.
- FR-26: The system shall record the offer amount and relevant offer information.
- FR-27: The stakeholder shall be able to view buyer offers.
- FR-28: The stakeholder shall be able to accept an offer.
- FR-29: The stakeholder shall be able to reject an offer.
- FR-30: The system shall update the offer status after an acceptance or rejection.

7.6 Orders and Payment

- FR-31: The system shall allow an eligible accepted transaction to proceed to an order.
- FR-32: The system shall record order information and order status.
- FR-33: The buyer shall be able to provide a delivery address for an order.
- FR-34: The buyer shall be able to make a payment through the approved payment process.
- FR-35: The system shall record relevant payment status/information.
- FR-36: The system shall prevent an order from being treated as paid unless the approved payment process confirms payment.

7.7 Delivery and Order Tracking

- FR-37: The stakeholder shall be able to update relevant delivery details.
- FR-38: The buyer shall be able to view the current order/delivery status.
- FR-39: The system shall provide order tracking information available from the agreed delivery process.

- FR-40: The buyer shall be able to confirm receipt of the livestock/order.

7.8 Ratings and Reviews
- FR-41: A buyer shall be able to rate a completed order.
- FR-42: A buyer shall be able to give a review for a completed order.
- FR-43: The system shall associate the rating/review with the relevant completed order and buyer.

7.9 Seller Financial Summary
- FR-44: The stakeholder shall be able to view a financial summary of relevant sales/transactions.
- FR-45: The financial summary shall use recorded transaction/payment information available to the system.

7.10 Market Access and Category Management
- FR-46: The platform shall make the stakeholder's livestock available to buyers.
- FR-47: The system shall support multiple livestock categories.
- FR-48: Livestock categories/types shall be manageable without changing the main trading workflow.

8. Non-Functional Requirements

8.1 Usability
- NFR-01: The app shall provide a simple and user-friendly interface.
- NFR-02: Important livestock and transaction information shall be clearly presented.
- NFR-03: Buyers should be able to move from browsing to filtering, viewing details and making an offer without unnecessary steps.
- NFR-04: Order, delivery and payment statuses should be easy to understand.

8.2 Performance
- NFR-05: Normal user actions should respond within a reasonable time under normal conditions.
- NFR-06: Listings and relevant details should load efficiently.
- NFR-07: Filtering should return relevant results without unnecessary delay.
- NFR-08: Order/status updates should be reflected without unnecessary delay.

8.3 Security
- NFR-09: Accounts shall use secure authentication.
- NFR-10: Role-based access shall restrict functions appropriately.
- NFR-11: Ownership documents shall be protected from unauthorised access.
- NFR-12: Payment-related information shall be protected and handled through the approved payment mechanism.
- NFR-13: Buyer, seller, livestock and transaction information shall be stored securely.

8.4 Availability and Reliability
- NFR-14: The application should be available when its supporting services are operational.
- NFR-15: Successful listing, offer, order and transaction submissions should be reliably stored.
- NFR-16: Stored livestock, offer, order and transaction information should be retrieved accurately.

8.5 Scalability
- NFR-17: The system should accommodate additional buyers and livestock listings.
- NFR-18: The system should provide a flexible foundation for multiple livestock categories.

- NFR-19: The database structure should support the additional Phase 2 trading entities without unnecessary duplication.

9. Integration Requirements

9.1 Flutter

- Provide application screens and user interaction.
- Provide buyer and stakeholder authentication screens.
- Display listings and detailed livestock information.
- Provide filtering, offer, order, delivery, payment, tracking and review interfaces.
- Provide seller listing, offer-response, delivery and financial-summary interfaces.

9.2 ASP.NET

- Provide the backend/API.
- Handle business logic and validation.
- Control role-based access.
- Process livestock, offer, order, delivery and review operations.
- Process or integrate with the approved payment mechanism.
- Provide data needed for seller financial summaries.

9.3 Supabase

- Provide authentication.
- Store application data through the database.
- Store livestock listings, offers, orders, delivery information, payment status and reviews as designed.
- Store ownership documentation and relevant media through approved storage mechanisms.

9.4 Integration Flow

Flutter Application → ASP.NET Backend/API → Supabase Services/Database → Approved External Payment/Delivery Services where required

The buyer or stakeholder interacts with Flutter. Requests are sent to the ASP.NET backend, which validates the request, applies business rules and communicates with Supabase and any approved external services. Results and status updates are returned to the Flutter application.

10. Phase 2 Trading Workflow

Step	Buyer	Stakeholder / System
1. Discover	Browse and filter livestock.	Listings are managed by stakeholder.
2. Evaluate	View livestock details, ownership/brandmark and vaccination information where provided.	Stakeholder maintains listing information and documents.
3. Offer	Submit an offer.	Stakeholder views and accepts/rejects offer.
4. Order	Place/confirm eligible order and provide delivery address.	System creates/updates order.

5. Payment	Complete approved payment.	System records payment status.
6. Delivery	View tracking/status.	Stakeholder updates relevant delivery details.
7. Completion	Confirm receipt.	System marks order as received/completed.
8. Feedback	Rate and review completed order.	System stores review against completed order.
9. Finance	—	Stakeholder views financial summary.

11. Requirements Change Log

Change	Reason	Effect on Requirements
Multiple sellers changed to one stakeholder/seller.	Mentor feedback to keep the project focused only on stakeholder and his needs.	Seller functionality is restricted to the stakeholder.
Buyer filtering added.	Buyers should be able to filter to their preferred livestock features.	Filtering by price, type and weight is now a functional requirement
International market requirements removed	Project scope was too broad; mentor feedback was to narrow it down to local.	focuses on local and national buyers and excludes international market functionality
Multiple livestock category scope removed.	Platform will be limited to cattle livestock as opposed to all livestock like pig/sheep.	Multiple livestock categories are excluded in the initial functionality.
Record-keeping narrowed to livestock information.	Avoid making the platform a broad marketplace	Animal information is linked to the livestock trading platform.
Phase 1 discovery expanded into transaction workflow.	Phase 2 continuation requires a complete trading process.	Offers, orders, delivery, payment, tracking and reviews added.
Buyer offers added.	Buyers need a way to make purchase offers.	FR-25 to FR-30 added.
Seller responses to offers added.	Stakeholder must control acceptance/rejection.	FR-27 to FR-30 added.
Orders added.	Accepted/eligible transactions need an order record.	FR-31 to FR-32 added.
Delivery address and tracking added.	Orders require delivery information and status visibility.	FR-33 and FR-37 to FR-40 added.
Payment added.	The expanded trading workflow requires payment.	FR-34 to FR-36 added; previous Phase 1 payment exclusion is superseded for Phase 2.

Receipt confirmation added.	Buyer needs to confirm completion.	FR-40 added.
Ratings and reviews added.	Provide post-purchase feedback.	FR-41 to FR-43 added.
Seller financial summary added.	Stakeholder needs visibility of sales/transaction results.	FR-44 to FR-45 added.
Seller listing management clarified.	Phase 2 requires full listing lifecycle.	Create, update and remove functions explicitly defined.

12. Requirements Traceability

Stakeholder Need	Requirement IDs	System Response
Improve market access.	FR-46 to FR-48	Stakeholder livestock is made available through a multi-category platform.
Buyers need to discover livestock.	FR-18 to FR-24	Buyers browse, view and filter listings.
Buyers need detailed information.	FR-17, FR-19	Relevant livestock details are displayed.
Buyers need to make offers.	FR-25 to FR-26	Buyer can submit an offer.
Stakeholder needs offer control.	FR-27 to FR-30	Stakeholder accepts/rejects offers.
Buyers need to complete purchases.	FR-31 to FR-36	Orders, delivery address and payment are supported.
Buyers need delivery visibility.	FR-37 to FR-40	Delivery details, tracking and receipt confirmation are supported.
Buyers need post-purchase feedback.	FR-41 to FR-43	Completed orders can be rated/reviewed.
Stakeholder needs transaction visibility.	FR-44 to FR-45	Financial summary is provided.

13. Acceptance Criteria – Phase 2

- The stakeholder can log in as the only seller
- The stakeholder can create, update and remove livestock listings.
- Buyers can browse and filter listings.
- Buyers can view relevant livestock details, including available ownership, brandmark and vaccination information.
- A buyer can make an offer and the stakeholder can accept or reject it.
- An eligible transaction can create an order.
- The buyer can enter a delivery address.

- The buyer can complete an approved payment.
- The buyer can view order/delivery status.
- The stakeholder can update delivery details.
- The buyer can confirm receipt.
- The buyer can rate and review a completed order.
- The stakeholder can view a financial summary.
- Role-based security prevents unauthorised access to seller and buyer functions.
- The system stores and retrieves Phase 2 data correctly.

14. Open / Unconfirmed Requirements

- The exact payment provider and payment implementation method must be confirmed.
- The exact delivery/tracking mechanism must be confirmed.
- The business rule determining when an accepted offer becomes an order must be confirmed.
- Whether ownership documents are visible directly to buyers or only shown as verified/available information must be confirmed.
- The exact financial-summary fields and calculations must be confirmed.
- The exact rating scale and review rules must be confirmed.
- Advanced auction/bidding remains outside scope unless separately approved.

16. Conclusion

Phase 2 continues the revised Phase 1 Commercial Livestock Trading Platform and expands it into a structured digital trading workflow. The stakeholder remains the only seller, while buyers can discover livestock, filter and inspect listings, make offers, complete eligible orders, provide delivery information, pay, track delivery, confirm receipt and provide ratings/reviews. The stakeholder can manage listings, respond to offers, manage delivery details and view a financial summary.

The updated requirements provide the basis for the Phase 2 system architecture, use case diagram, sequence diagrams, ERD and interface designs. All Phase 2 design artefacts should remain traceable to these requirements and to any further requirements formally confirmed by the stakeholder or mentor.

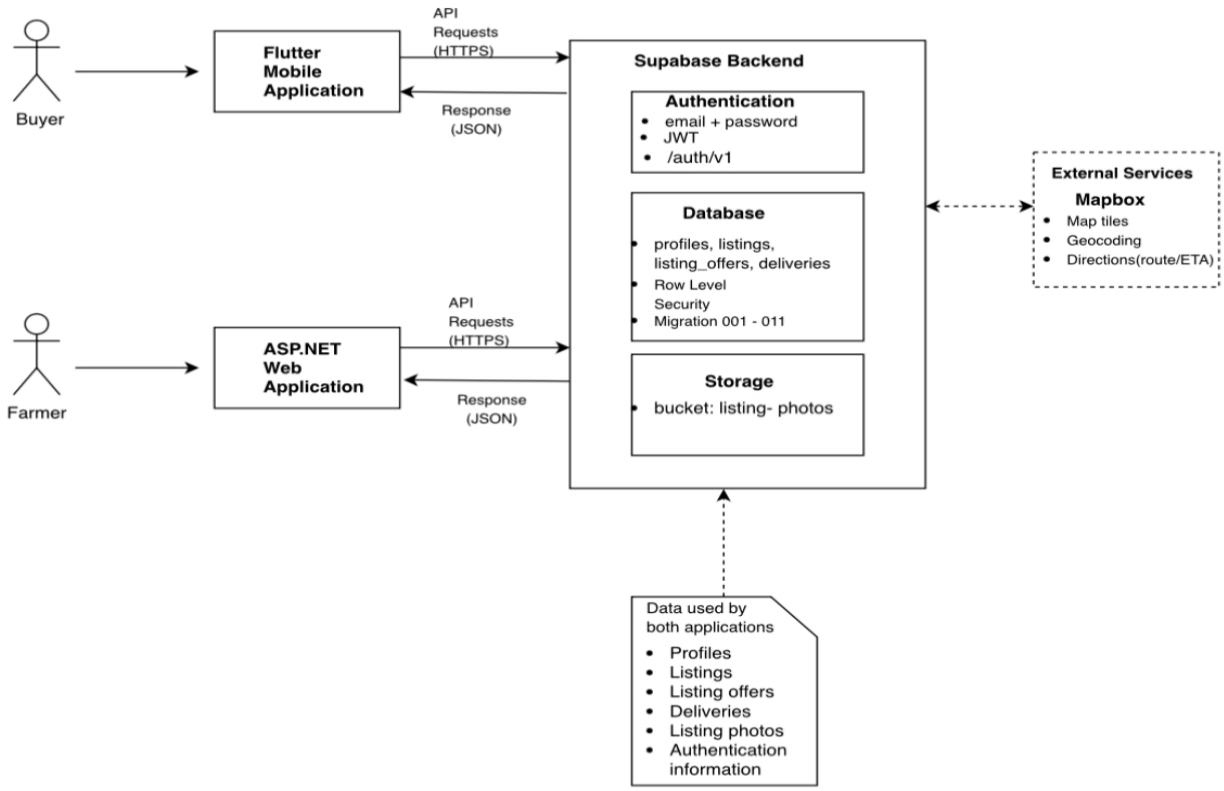
End of revised phase 1

SYSTEM ARCHITECTURE AND DESIGN – PHASE 2

1. SYSTEM ARCHITECTURE DESIGN

The AgriCast system consists of a Flutter mobile application, an ASP.NET web application and a shared Supabase backend. The Flutter application provides the mobile interface for system users, while the ASP.NET web application provides the web-based functionality. Both applications communicate with the shared Supabase backend, which manages the system's database and other relevant backend services. This architecture allows the applications to operate as an integrated system while using a common source of data.

1. Users / roles



The architecture shows who interacts with the system, such as:

- Buyer
- Farmer
- Administrator

2. Flutter Mobile Application

Flutter app handles the mobile-side functions, such as users viewing/interacting with livestock listings. The Flutter mobile application is used by farmers and buyers. Farmers can create and manage livestock listings, while buyers can browse listings, make offers, purchase livestock, and manage deliveries. The document identifies the farmer and buyer roles explicitly.

3. ASP.NET Web Application

The ASP.NET Core Web API serves as the application/business-logic layer between the Flutter application and Supabase. It handles authentication, listings, offers, payments, finance, deliveries and uploads through its API controllers. The architecture shows what the web application is responsible for, such as administrative/web management functions.

4. Supabase Backend

Supabase provides the shared backend services, such as Authentication using email/password and JWT, PostgreSQL database for profiles, listings, offers and deliveries; PostgREST for database access; Row Level Security (RLS) for controlling data ownership; Storage for listing photographs; and Database triggers, such as automatically marking a listing as sold after an offer is paid. So the architecture shows a shared backend and the relevant services the project actually uses, such as:

- Supabase Database
- Authentication
- Storage

5. Connections/data flow

FLUTTER sends authenticated HTTPS/JSON requests containing a Bearer token to the ASP.NET Core Web API. ASP.NET Core then communicates with Supabase on Flutter's behalf. The document explicitly states that the app never talks to Supabase directly. The arrows show how information moves between the applications and Supabase.
ASP.NET Core communicates with Supabase through the following services:

/auth/v1 for authentication
/rest/v1 for database operations
/storage/v1 for file storage

The ASP.NET Core services use the user's authentication token when accessing Supabase rather than exposing the service-role key.

FORMATION OF ONE INTEGRATED SOLUTION

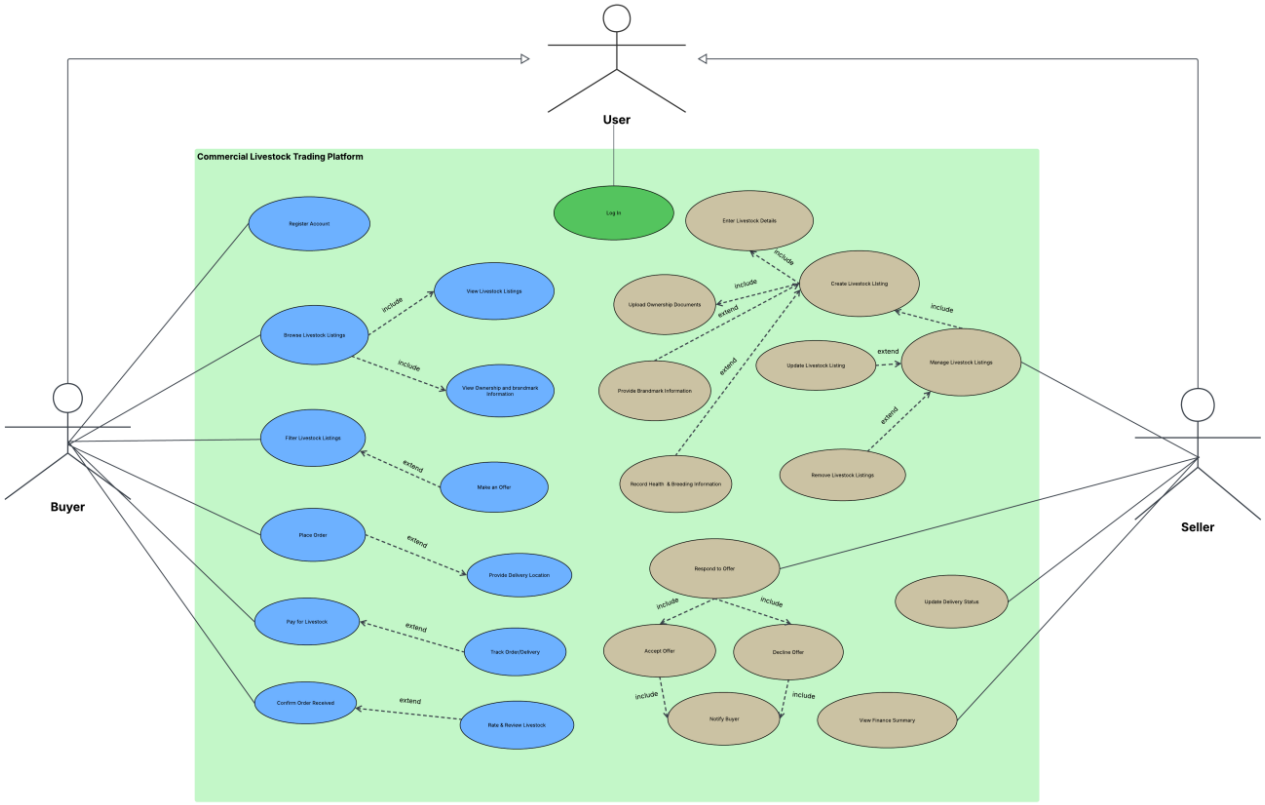
Flutter provides the mobile user interface and device functionality, while ASP.NET Core provides the web/API and business-logic layer. Supabase provides the centralised backend, authentication, database and storage used by the system. Therefore, the applications work together as one system rather than as separate applications: Flutter → ASP.NET Core → Supabase. Mapbox supplements the solution by providing mapping, geocoding and route/ETA functionality.

System Architecture Summary

The AgriCast system integrates a **Flutter mobile application**, an **ASP.NET web application**, and a shared **Supabase backend**. The Flutter application provides the mobile interface for users, while the ASP.NET application provides the web-based functionality. Both applications communicate with Supabase to access and manage the system's data and relevant backend services. This integrated architecture enables the different components to work together as a single system.

2. USE CASE DIAGRAM

The Use Case Diagram illustrates the main actors who interact with the AgriCast system and the major functions available to them. It defines the system boundary and shows the relationships between the actors and the system use cases. The diagram is based on the approved Phase 1 requirements and represents the proposed functionality of the AgriCast system.



The Use Case Diagram shows the main user roles and their interactions with the AgriCast system. The AgriCast farmer and buyers interact with the system through the relevant mobile functionality, while administrators interact with the system through the web application. The use cases represent the main functions required by the approved system scope.

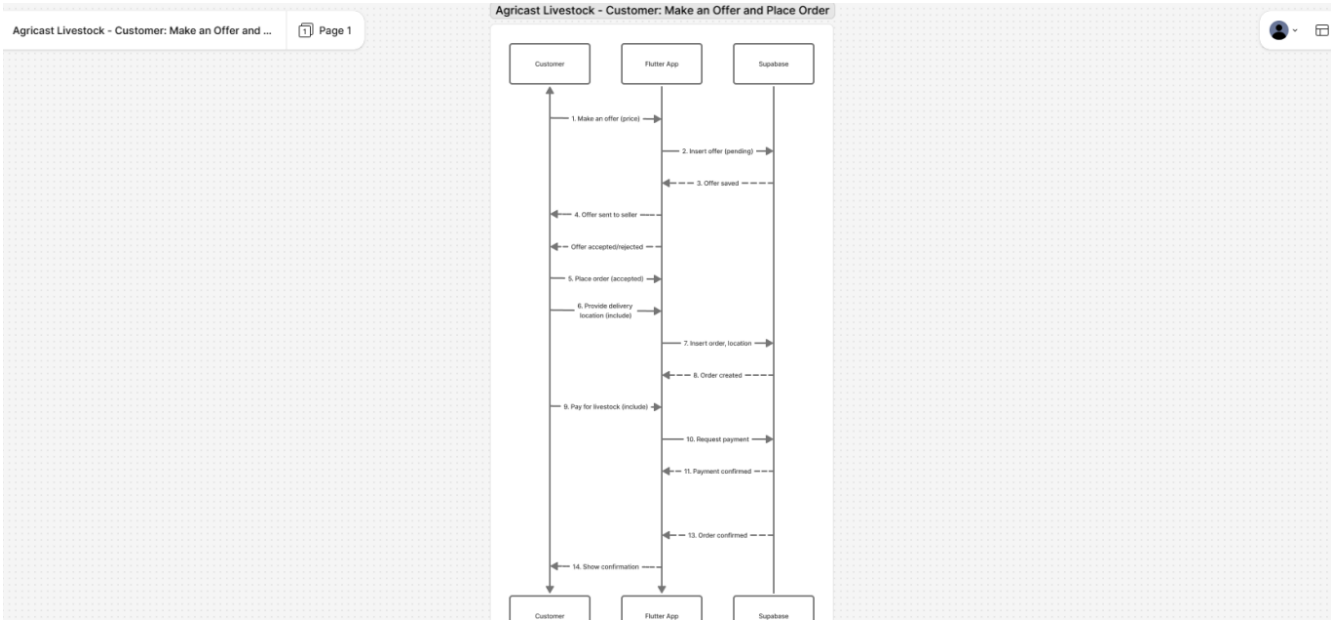
Actor	Main interactions
Farmer	Manages and views livestock listings and related information
Buyer	Views, filters and buys livestock listings
Administrator	Manages and oversees system information

3. SEQUENCE DIAGRAMS

Sequence diagrams illustrate the interaction between users, the application components, ASP.NET and the Supabase backend during important system processes. The diagrams show the order of requests and responses between the participants and are aligned with the approved use cases and requirements.

Sequence Diagram 1 — Flutter + Supabase

Buyer views/searches livestock listings
Buyer → Flutter App → Supabase → Flutter App → Buyer



The above sequence explained:

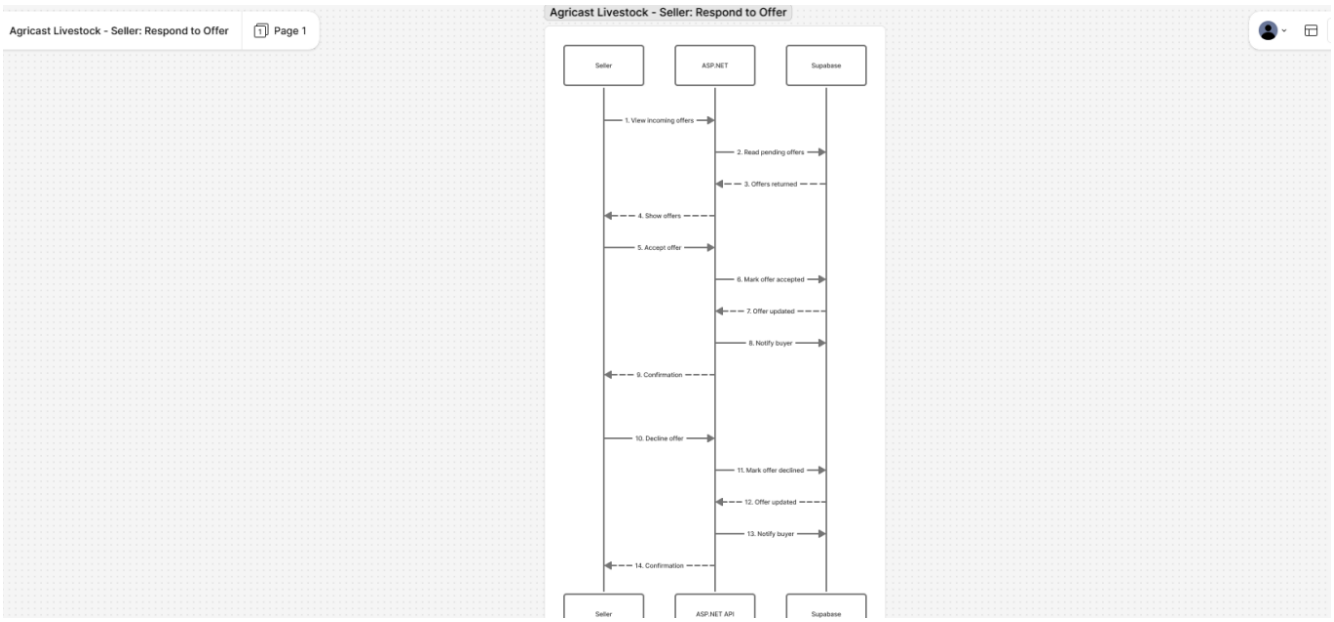
1. Buyer makes an offer.
2. The offer can be rejected or accepted.
3. Place order upon acceptance of offer.
4. Supabase returns the data to Flutter.
5. Flutter displays the listings to the buyer.

This sequence diagram illustrates how a buyer uses the Flutter mobile application to view livestock listings. The Flutter application sends a request to Supabase to retrieve the relevant livestock listing information. Supabase processes the request and returns the data to the application, which then displays the available listings to the buyer.

Sequence Diagram 2 — ASP.NET + Supabase

Admin reviews/manages livestock listing

Admin → ASP.NET Web App → Supabase → ASP.NET Web App → Admin



The above sequence explained:

1. Admin logs into the ASP.NET web application.
2. ASP.NET sends a request to Supabase.
3. Supabase retrieves the listing information.
4. Supabase returns the information to ASP.NET.
5. ASP.NET displays the listing to the admin.
6. Admin performs an action, such as updating the listing/status.
7. ASP.NET sends the update to Supabase.
8. Supabase saves the changes.
9. ASP.NET displays confirmation.

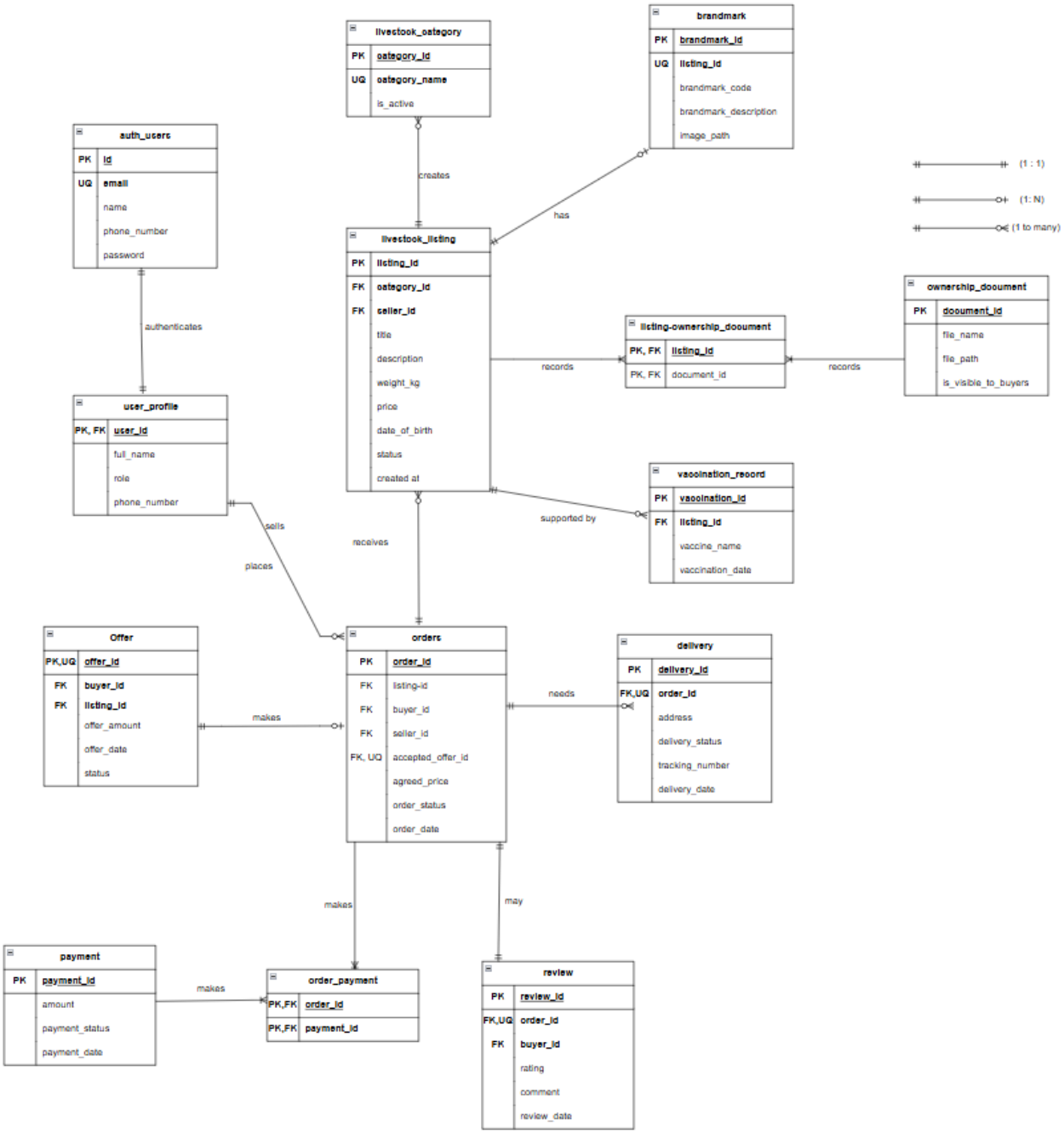
This sequence diagram illustrates how an administrator uses the ASP.NET web application to manage livestock listing information. The ASP.NET application communicates with Supabase to retrieve and update the relevant listing data. Supabase processes the request and returns the appropriate response to the ASP.NET application, which displays the updated information to the administrator.

4. DATABASE DESIGN: ENTITY RELATIONSHIP DIAGRAM (ERD)

Purpose and Design Approach

This section shows how the information required by the approved system will be organised in the Supabase
database (PostgreSQL). It is based only on the approved functional requirements (FR-01 to FR-29), the non-functional requirements (NFR-01 to NFR-17), the scope and exclusions in Section 5, and the integration flow (Flutter → ASP.NET → Supabase).

The diagram below is an Entity Relationship Diagram for the Commercial Livestock Trading Platform.



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ENTITIES AND ATTRIBUTES

The Entity Relationship Diagram (ERD) illustrates the structure of the Agricast Advisory livestock marketplace database, which will be implemented using Supabase. It identifies the entities, their attributes, primary keys, foreign keys, and relationships required to support the proposed Flutter mobile application and ASP.NET web application. The database is designed to manage user accounts, livestock listings, ownership documentation, vaccination records, offers, orders, payments, deliveries, and customer reviews.

-auth_users: Stores user authentication information, including the user's ID, email, name, phone number, and password-related information.

- user_profile: Stores additional user information, including the user's full name, role, and phone number. It is linked to the authentication entity.

- livestock_category: Stores livestock categories, including the category name and active status.

- livestock_listing: Stores livestock details, including the category, seller, title, description, weight, price, date of birth, status, and creation date.

- brandmark: Stores brandmark information associated with livestock listings, including the brandmark code, description, and image path.

- ownership_document: Stores ownership documentation details, including the file name, file path, and visibility to buyers.

- listing_ownership_document: Acts as a junction table linking livestock listings to ownership documents and supports the many-to-many relationship between them.

- vaccination_record:Stores livestock vaccination information, including the vaccination name and vaccination date.

- offer: Records offers submitted by buyers for livestock listings, including the buyer, listing, offer amount, offer date, and status.

- orders: Stores transaction details, including the livestock listing, buyer, seller, accepted offer, agreed price, order status, and order date.

- delivery:Stores delivery information associated with orders, including the delivery address, delivery status, tracking number, and delivery date.

- payment:Stores payment details, including the amount, payment status, and payment date.

- order_payment:Acts as a junction table linking orders and payments, allowing the database to represent the many-to-many relationship between these entities.

- review:Stores buyer reviews of completed transactions, including the rating, comment, and review date.

Primary Keys and Foreign Keys

Each entity has a primary key that uniquely identifies its records. Foreign keys connect related entities and maintain referential integrity. For example, the livestock_listing table contains foreign keys referencing livestock_category and user_profile, while the orders table references the relevant listing,

buyer, seller, and accepted offer. The listing_ownership_document and order_payment junction tables use composite primary keys consisting of their respective foreign keys to prevent duplicate associations.

RELATIONSHIP AND CARDINALITY

1.Auth Users and User Profile (One-to-One):Each authenticated user is associated with a corresponding user profile.

2.User Profile and Livestock Listing (One-to-Many):A seller can create multiple livestock listings, while each listing is associated with one seller.

3.Livestock Category and Livestock Listing (One-to-Many):A category can contain multiple livestock listings, while each listing belongs to one category.

4.Livestock Listing and Brandmark (One-to-Many):A livestock listing can have multiple brandmark records, while each brandmark record belongs to a listing.

5.Livestock Listing and Ownership Document (Many-to-Many): A listing can be associated with multiple ownership documents, and a document can be associated with multiple listings. This relationship is implemented through the listing_ownership_document junction table.

6.Livestock Listing and Vaccination Record (One-to-Many): A listing can have multiple vaccination records, while each vaccination record belongs to one listing.

7.User Profile and Orders (One-to-Many): A user can participate in multiple orders in the role of buyer or seller. Each order references one buyer and one seller.

8.Livestock Listing and Orders (One-to-Many): A livestock listing can be associated with multiple order records over time, while each order references one listing.

9.User Profile and Offers (One-to-Many): A buyer can submit multiple offers, while each offer is associated with one buyer and one livestock listing.

10.Offer and Orders (One-to-Many): An offer may be referenced by multiple orders in the current diagram, while each order references an accepted offer. The system rules should determine whether an offer can be accepted for more than one order.

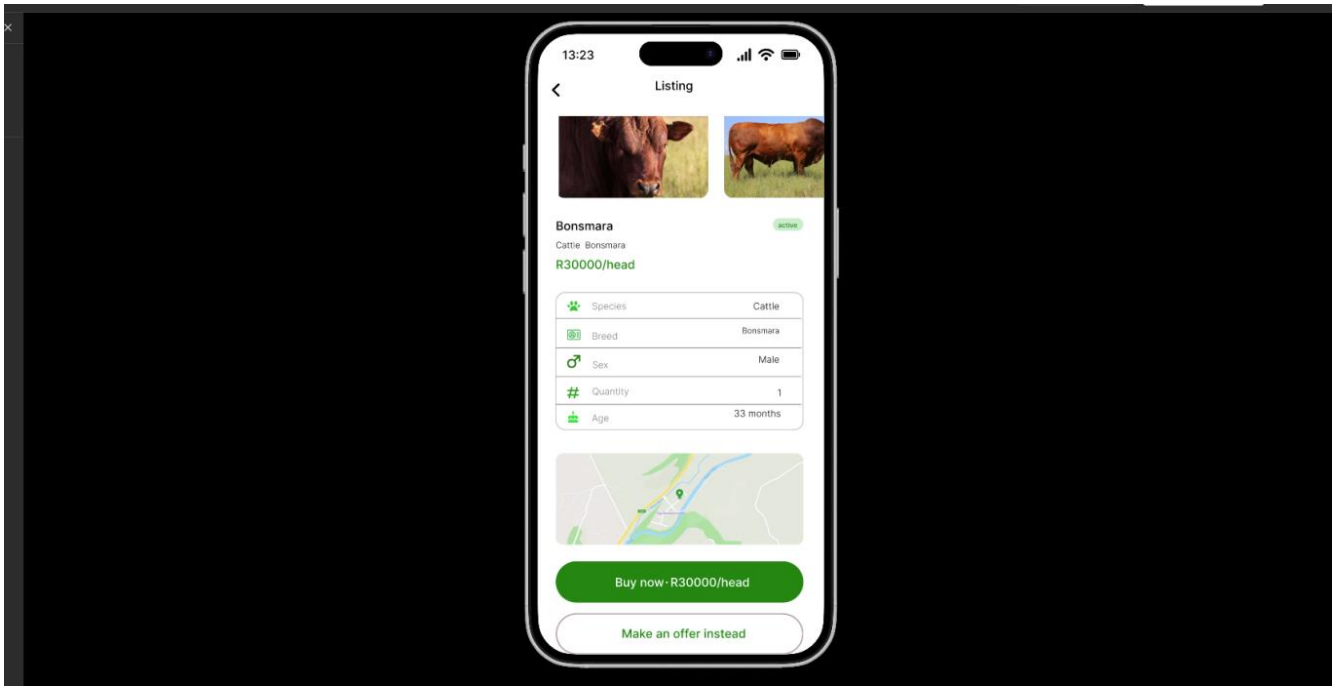
11.Orders and Delivery (One-to-One): Each order is associated with one delivery record, and each delivery record belongs to one order.

12.Orders and Payment (Many-to-Many): An order can be associated with multiple payments, and a payment can be associated with multiple orders. This relationship is implemented through the order_payment junction table.

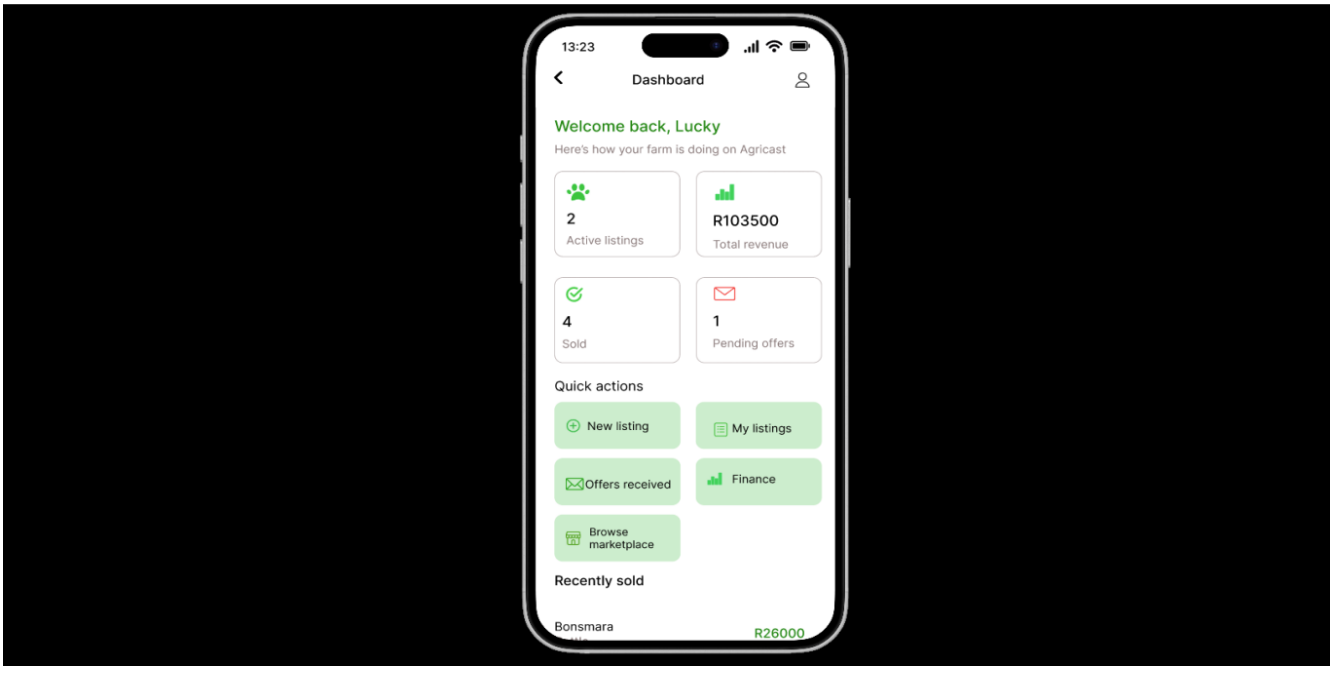
13.Orders and Reviews (One-to-Many): An order can have multiple reviews, while each review belongs to one order and is submitted by a buyer.

Our ERD supports the core functions of the Agricast-advisory livestock marketplace by organising livestock information, facilitating access to the market, recording buyer offers and orders, maintaining ownership and vaccination records, and supporting payment, delivery, and review management. The primary keys, foreign keys, and junction tables help maintain data consistency and establish connections between the information managed by the Flutter mobile application and the ASP.NET web application. Overall, the database design provides a structured foundation for implementing the approved system requirements in Supabase.

5. FLUTTER DESIGN



6. ASP.NET DESIGN



7. DESIGN ALIGNMENT TABLE

The following table is the design alignment process that demonstrates how each approved system requirement has been translated into corresponding system design artefacts. Each requirement is represented through an appropriate use case that defines the required system functionality, a sequence diagram that illustrates the interactions and flow of the associated process, and relevant database entities that provide the required data structure. The requirement is also linked to the corresponding user interface through which the functionality is accessed and performed.

This alignment ensures consistency between the approved requirements and the proposed system architecture, UML diagrams, database design, and user interface designs. It provides a clear traceability relationship between the functional requirements and their implementation within the proposed system, ensuring that each design element has a defined purpose and is supported by the overall system requirements.

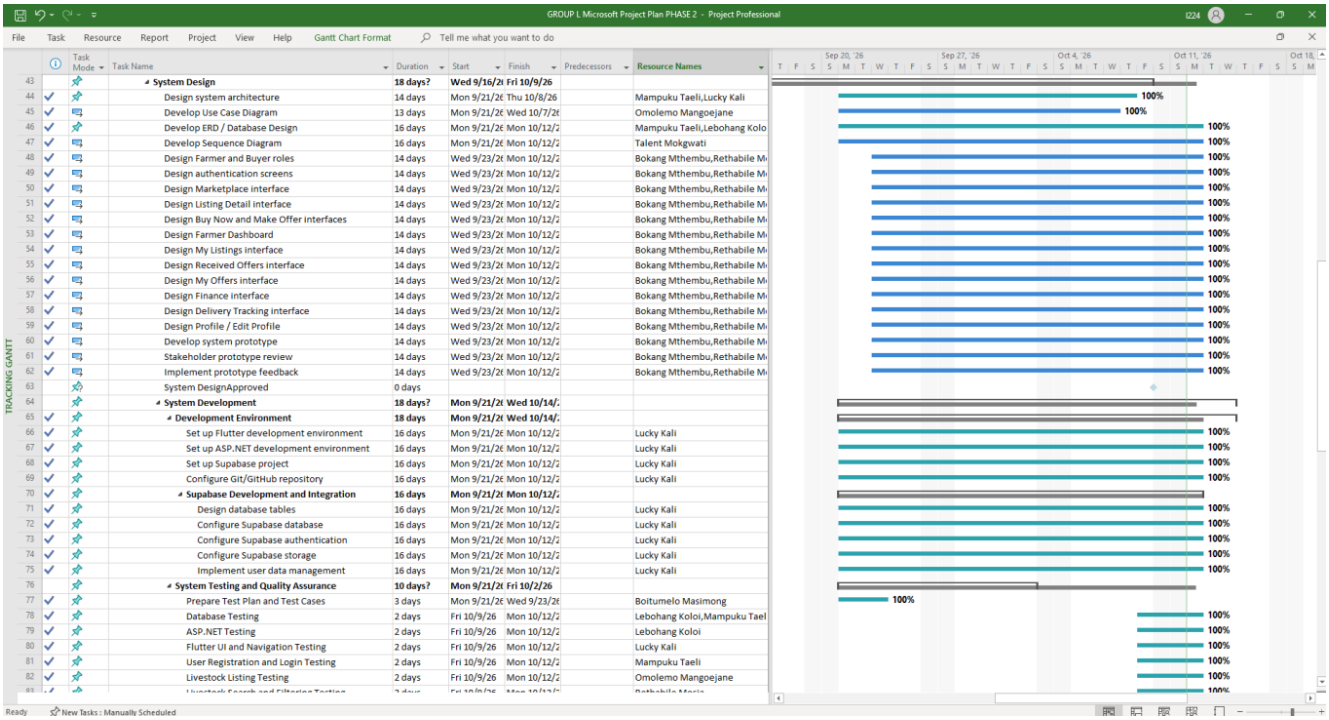
Requirement	Use Case	Sequence Diagram	ERD Element	Interface
Buyer can register and log in	Register / Log In	Authentication	Auth Users, User Profile	Flutter – Login/Register Screen
Stakeholder can manage livestock listings	Manage Livestock Listings	Sequence Diagram 1	User Profile, Livestock Listing	Flutter – Livestock Management Screen
Stakeholder can manage livestock information	Manage Livestock Details	Sequence Diagram 1	Livestock Listing, Brandmark, Ownership Document, Vaccination Record	Flutter – Livestock Details Screen
Buyer can view livestock listings	View Livestock Listings	Sequence Diagram 1	Livestock Listing, Livestock Category	Flutter – Marketplace Screen
Buyer can view detailed livestock information	View Livestock Details	Sequence Diagram 1	Livestock Listing, Brandmark, Ownership Document, Vaccination Record	Flutter – Livestock Details Screen
Buyer can filter livestock listings	Filter Livestock Listings	Sequence Diagram 1	Livestock Listing, Livestock Category	Flutter – Marketplace/Filter Screen
Buyer can make an offer	Make Offer	Sequence Diagram 1	Offer, User Profile, Livestock Listing	Flutter – Offer Screen
Stakeholder can accept or reject offers	Manage Offers	Sequence Diagram 2	Offer, Livestock Listing	Flutter – Seller Offers Screen
Buyer can place an eligible order	Place Order	Sequence Diagram 1	Orders, Offer, Livestock Listing, User Profile	Flutter – Checkout/Orders Screen
Buyer can provide a delivery address	Provide Delivery Details	Sequence Diagram 1	Delivery, Orders	Flutter – Delivery Address Screen

Buyer can make a payment	Process Payment	Sequence Diagram 1	Payment, Order Payment, Orders	Flutter – Payment Screen
System can record payment status	Manage Payment Status	Sequence Diagram 1	Payment, Order Payment, Orders	Flutter – Payment/Order Status Screen
Stakeholder can manage delivery details	Manage Delivery	Sequence Diagram 2	Delivery, Orders	Flutter – Seller Delivery Management Screen
Buyer can track an order	Track Order	Sequence Diagram 1	Delivery, Orders	Flutter – Order Tracking Screen
Buyer can confirm receipt of livestock	Confirm Receipt	Sequence Diagram 1	Orders, Delivery	Flutter – Order Details Screen
Buyer can rate and review a completed order	Rate and Review Order	Sequence Diagram 1	Review, Orders, User Profile	Flutter – Ratings and Reviews Screen
Stakeholder can view a financial summary	View Financial Summary	Sequence Diagram 2	Orders, Payment	Flutter – Seller Finance Screen
Platform supports multiple livestock categories	Manage/View Livestock Categories	Sequence Diagram 1	Livestock Category, Livestock Listing	Flutter – Marketplace Screen
System provides secure role-based access	Manage User Access	Authentication	Auth Users, User Profile	Flutter – Login Screen; ASP.NET – Access Control
Administrator can manage system information	Manage System Information	Sequence Diagram 2	Livestock Listing, User Profile	ASP.NET – Admin Dashboard

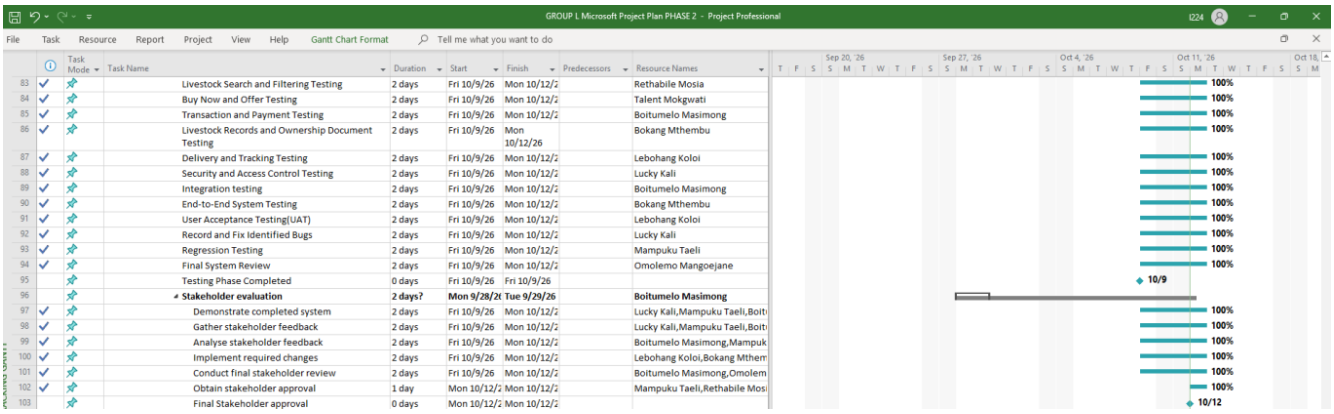
DESIGN SOURCE LINKS

Design	Tool Used	Source Link / Editable File Name
System Architecture Diagram	draw.io	*https://tinyurl.com/systemArch*
Use Case Diagram	LucidSpark	*https://lucid.app/lucidspark/useCaseDiagram*
Sequence Diagrams *(Both)*	Figma	*https://www.figma.com/board/CustomerSequence* *https://www.figma.com/SellerSequencce*
Entity Relationship Diagram (ERD)	SQL Data Modeler	*https://tinyurl.com/entityRelatedDiagram*
Flutter Wireframes	Figma	*https://www.figma.com/proto/MobileApp*
ASP.NET Wireframes	Figma	*https://www.figma.com/proto/WebApp*
Navigation Flow	Figma	*https://www.figma.com/boardNavFlow*

MICROSOFT PROJECT PLAN



	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
43		**System Design**	**18 days?**	**Wed 9/16/2**	**Fri 10/9/26**		
44		Design system architecture	14 days	Mon 9/21/2	Thu 10/8/26		Mampuku Taeli,Lucky Kali
45		Develop Use Case Diagram	13 days	Mon 9/21/2	Wed 10/7/2		Omolemo Mangoejane
46		Develop ERD / Database Design	16 days	Mon 9/21/2	Mon 10/12/		Mampuku Taeli,Lebohang Kolo
47		Develop Sequence Diagram	16 days	Mon 9/21/2	Mon 10/12/		Talent Mokgwati
48		Design Farmer and Buyer roles	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
49		Design authentication screens	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
50		Design Marketplace interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
51		Design Listing Detail interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
52		Design Buy Now and Make Offer interfaces	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
53		Design Farmer Dashboard	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
54		Design My Listings interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
55		Design Received Offers interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
56		Design My Offers interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
57		Design Finance interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
58		Design Delivery Tracking interface	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
59		Design Profile / Edit Profile	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
60		Develop system prototype	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
61		Stakeholder prototype review	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
62		Implement prototype feedback	14 days	Wed 9/23/2	Mon 10/12/		Bokang Mthembu,Rethabile M
63		System DesignApproved	0 days				
64		**System Development**	**18 days?**	**Mon 9/21/2**	**Wed 10/14/**		
65		**Development Environment**	**18 days**	**Mon 9/21/2**	**Wed 10/14/**		
66		Set up Flutter development environment	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
67		Set up ASP.NET development environment	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
68		Set up Supabase project	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
69		Configure Git/GitHub repository	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
70		**Supabase Development and Integration**	**16 days**	**Mon 9/21/2**	**Mon 10/12/**		
71		Design database tables	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
72		Configure Supabase database	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
73		Configure Supabase authentication	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
74		Configure Supabase storage	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
75		Implement user data management	16 days	Mon 9/21/2	Mon 10/12/		Lucky Kali
76		**System Testing and Quality Assurance**	**10 days?**	**Mon 9/21/2**	**Fri 10/2/26**		
77		Prepare Test Plan and Test Cases	3 days	Mon 9/21/2	Wed 9/23/2		Boitumelo Masimong
78		Database Testing	2 days	Fri 10/9/26	Mon 10/12/		Lebohang Koloi,Mampuku Tael
79		ASP.NET Testing	2 days	Fri 10/9/26	Mon 10/12/		Lebohang Koloi
80		Flutter UI and Navigation Testing	2 days	Fri 10/9/26	Mon 10/12/		Lucky Kali
81		User Registration and Login Testing	2 days	Fri 10/9/26	Mon 10/12/		Mampuku Taeli
82		Livestock Listing Testing	2 days	Fri 10/9/26	Mon 10/12/		Omolemo Mangoejane



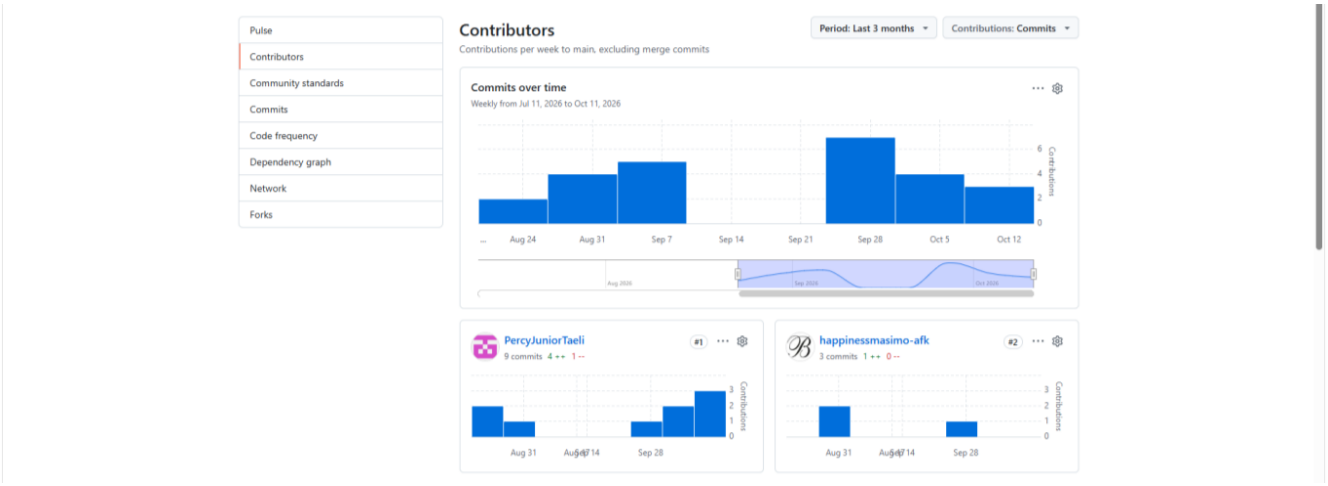
	Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
83		Livestock Search and Filtering Testing	2 days	Fri 10/9/26	Mon 10/12/		Rethabile Mosia
84		Buy Now and Offer Testing	2 days	Fri 10/9/26	Mon 10/12/		Talent Mokgwati
85		Transaction and Payment Testing	2 days	Fri 10/9/26	Mon 10/12/		Boitumelo Masimong
86		Livestock Records and Ownership Document Testing	2 days	Fri 10/9/26	Mon 10/12/26		Bokang Mthembu
87		Delivery and Tracking Testing	2 days	Fri 10/9/26	Mon 10/12/		Lebohang Koloi
88		Security and Access Control Testing	2 days	Fri 10/9/26	Mon 10/12/		Lucky Kali
89		Integration testing	2 days	Fri 10/9/26	Mon 10/12/		Boitumelo Masimong
90		End-to-End System Testing	2 days	Fri 10/9/26	Mon 10/12/		Bokang Mthembu
91		User Acceptance Testing(UAT)	2 days	Fri 10/9/26	Mon 10/12/		Lebohang Koloi
92		Record and Fix Identified Bugs	2 days	Fri 10/9/26	Mon 10/12/		Lucky Kali
93		Regression Testing	2 days	Fri 10/9/26	Mon 10/12/		Mampuku Taeli
94		Final System Review	2 days	Fri 10/9/26	Mon 10/12/		Omolemo Mangoejane
95		Testing Phase Completed	0 days	Fri 10/9/26	Fri 10/9/26		
96		**Stakeholder evaluation**	**2 days?**	**Mon 9/28/2**	**Tue 9/29/26**		**Boitumelo Masimong**
97		Demonstrate completed system	2 days	Fri 10/9/26	Mon 10/12/		Lucky Kali,Mampuku Taeli,Boit
98		Gather stakeholder feedback	2 days	Fri 10/9/26	Mon 10/12/		Lucky Kali,Mampuku Taeli,Boit
99		Analyse stakeholder feedback	2 days	Fri 10/9/26	Mon 10/12/		Boitumelo Masimong,Mampuk
100		Implement required changes	2 days	Fri 10/9/26	Mon 10/12/		Lebohang Koloi,Bokang Mthem
101		Conduct final stakeholder review	2 days	Fri 10/9/26	Mon 10/12/		Boitumelo Masimong,Omolem
102		Obtain stakeholder approval	1 day	Mon 10/12/	Mon 10/12/		Mampuku Taeli,Rethabile Mos
103		Final Stakeholder approval	0 days	Mon 10/12/	Mon 10/12/		

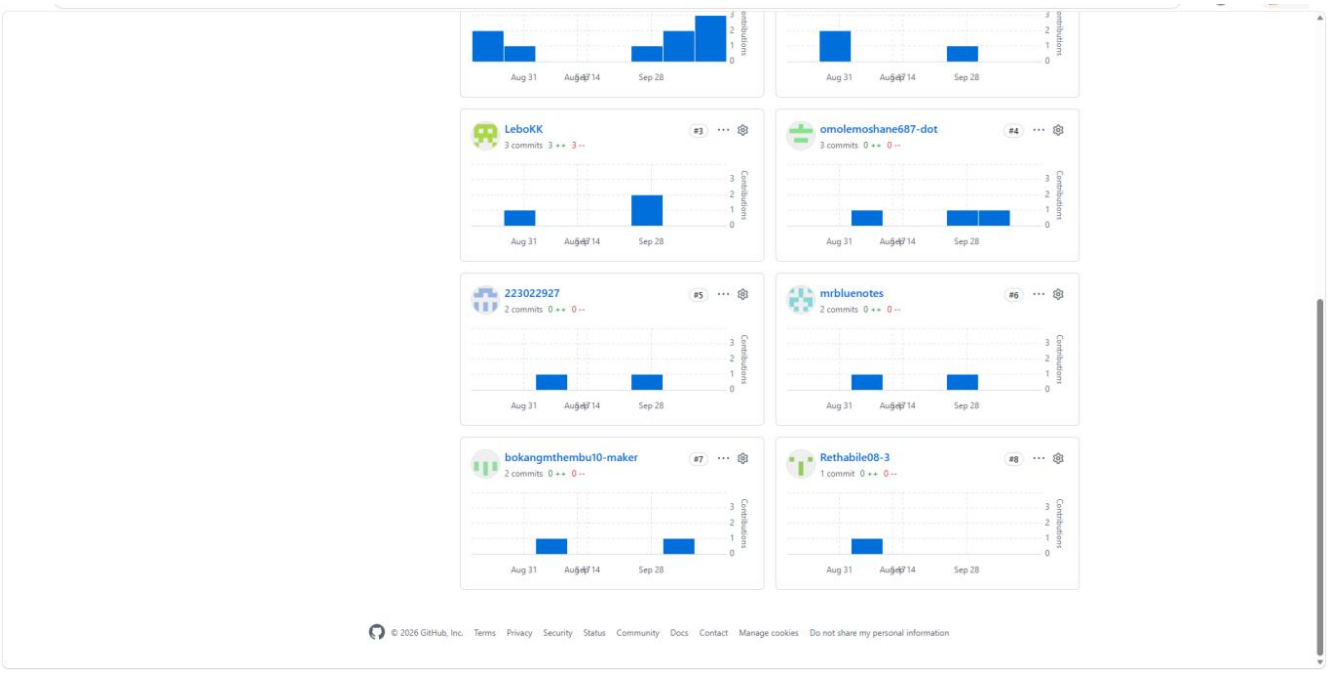
8. PROJECT PROGRESS TRACKER

	A	B	C	D	E	F	G	H	I	J
3	Task ID	Design / Project Item	Description / Expected Output	Responsible Student(s)	Planned Date	Actual Completion Date	Status	% Complete	Evidence / Link	Notes / Action Required
4	P2-01	System Architecture	Overall architecture showing Flutter, ASP.NET and shared Supabase backend	Mampuku Taeli	21/09/26	08/10/2026	Completed	100%	https://tinyurl.com/systemArch	
5	P2-02	Use Case Diagram	Actors, use cases, system boundary and appropriate UML relationships	Omolemo Mangoejane	21/09/26	07/10/2026	Completed	100%	https://lucid.app/lucidspark/useCaseDiagram	
6	P2-03	Sequence Diagram 1 (Buyer)	Important Flutter process	Talente Mokgwati	21/09/26	09/10/2026	Completed	100%	https://www.figma.com/board/CustomerSequence	
7	P2-04	Sequence Diagram 2(Seller)	Important ASP.NET process	Talente Mokgwati		12/10/2026	Completed	100%	https://www.figma.com/SellerSequencce	
8	P2-05	ERD	Entities, attributes, PKs, FKs, relationships and cardinality	Mampuku Taeli & Lebohang Koloi	21/09/26	12/10/2026	Completed	100%	https://tinyurl.com/entityRelatedDiagram	
9	P2-06	Flutter Interface Designs	Main mobile screens and navigation	Bokang Mthembu & Rethabile Mosia	21/09/26	10/10/2026	Completed	100%	https://tinyurl.com/entityRelatedDiagram	
10	P2-07	ASP.NET Interface Designs	Main web pages and navigation	Bokang Mthembu,Rethabile Mosia,Talente Mokgwati & Lucky Kali	21/09/26	12/10/2026	Completed	100%	https://www.figma.com/webApp	
11	P2-08	Design Alignment	Requirements linked across UML, ERD and interfaces	Lebohang Koloi,Mampuku Taeli & Boitumelo Masimong	21/09/26	08/10/2026	Completed	100%	Already provided	
12	P2-09	Updated SRS	Phase 2 architecture and design section inserted in existing SRS	Omolemo Mangoejane & Lebohang Koloi	21/09/26	13/10/2026	Completed	100%	In the zip folder	
13	P2-10	Microsoft Project	Schedule updated to reflect design progress and upcoming development	Mampuku Taeli	21/09/26	30/09/2026	Completed	100%	In the zip folder	
14	P2-11	GitHub	Current design documentation and repository organisation	All Members	21/09/26	13/10/2026	Completed	100%	https://github.com/happinessmasimo-afk/Tech_Elites	
15	P2-12	Phase 2 Presentation	Presentation and defence preparation	All Members	21/09/26	15/09/2026	Not Started	0%	N/A	
16										

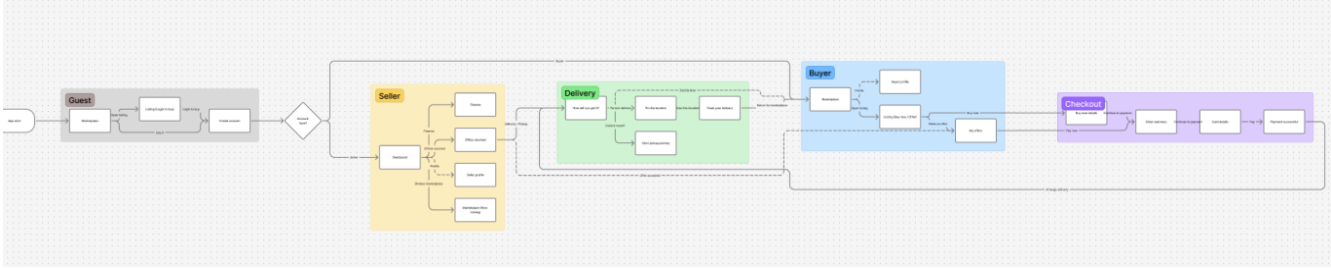
GITHUB

[GitHub Link]





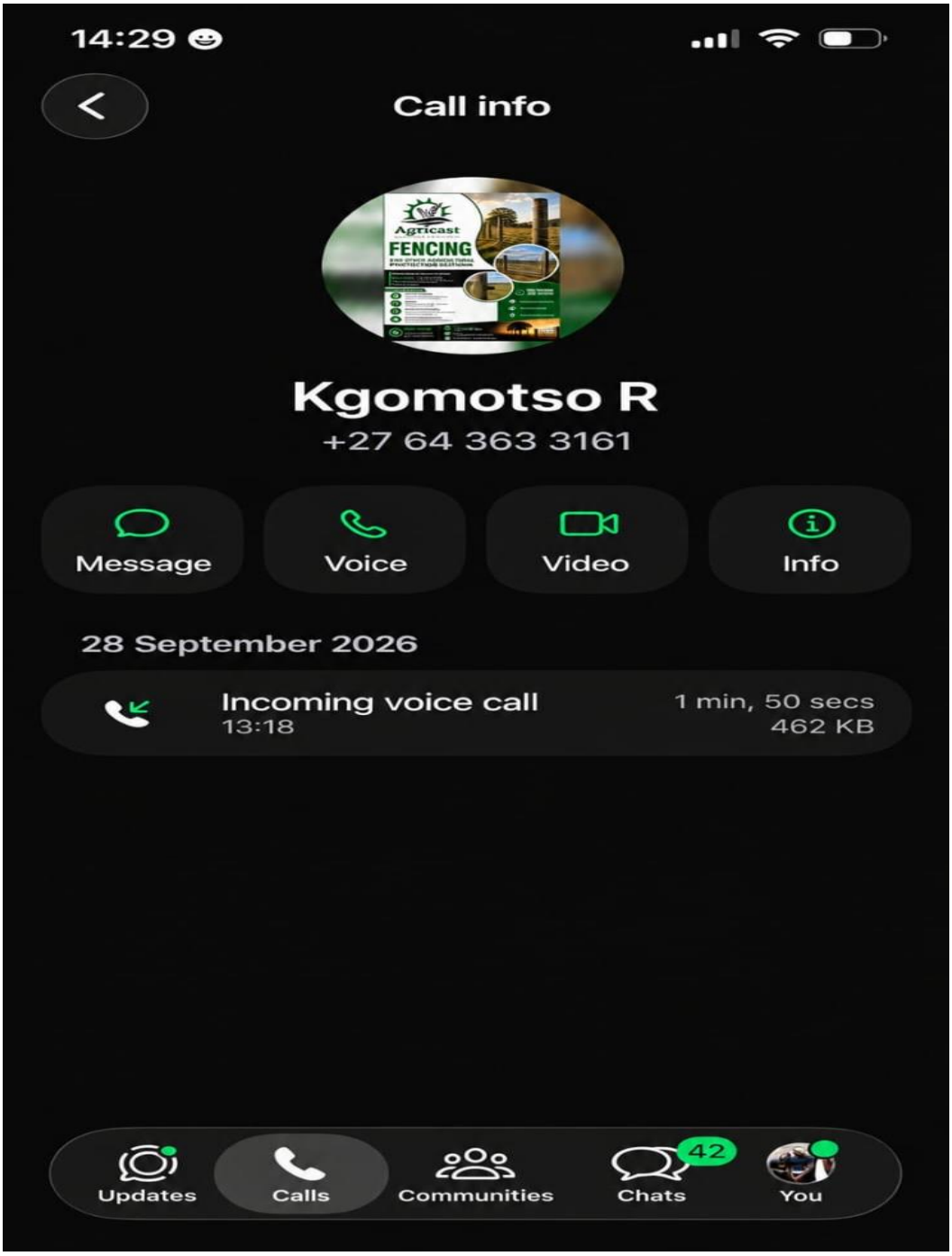
NAVIGATION FLOW



STAKEHOLDER FEEDBACK

During Phase 2, our group discussed the stakeholder's feedback on the revised AgriCast system requirements and designs on the below attached call log, including whether the platform meets the needs of both buyers and the seller. He confirmed the buyer functions, such as making offers, purchasing livestock, adding delivery addresses, making payments, tracking deliveries, and for buyers to leave ratings and reviews. We have also discussed his ability to manage listings, accept or reject offers, upload ownership documents, and update delivery details and if it was okay for us to incorporate it into the system we're building for him.

Finally, we clarified any changes needed to the Flutter and ASP.NET designs, database structure, and overall system functionality before proceeding with implementation of the Agricast livestock trading platform.


14:29
Call info
Agricast
FENCING
Kgomotso R
+27 64 363 3161
Message
Voice
Video
Info
28 September 2026
Incoming voice call
13:18
1 min, 50 secs
462 KB
Updates
Calls
Communities
Chats
42
You